PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	EFFECTIVENESS OF EXERCISE FOR OSTEOSARCOPENIA IN OLDER ADULTS: A SYSTEMATIC REVIEW PROTOCOL
AUTHORS	Vinícius-Souza, Guilherme; Noll, Matias; Silveira, Erika

VERSION 1 – REVIEW

REVIEWER	Locquet, Médéa University of Liege
REVIEW RETURNED	16-Nov-2020

GENERAL COMMENTS	The theme is new and important in terms of public health. However, the research protocol presented here needs further refinement. Title - The World Health Organization recommends that the term 'older adults' be preferred to 'elderly'. Abstract - "Improve the osteosarcopenia" does not sound right. - Page 6, line 3 : osteosarcopenia is misspelled Introduction - “Osteosarcopenia is a recent syndrome”, it seems to me rather that the syndrome has been recently defined - It would be interesting to identify some preliminary studies affirming that physical exercise could have a beneficial effect in the fight against osteosarcopenia. Methods - The definitions of osteoporosis/osteopenia and sarcopenia deserve further clarification. With regard to muscle health, do the authors consider, in the definition of osteosarcopenia, all individuals from the detection of osteopenia or will they make a distinction and a sensitivity analysis with a definition of osteosarcopenia involving all osteopenics and a definition of osteosarcopenia involving only osteoporotics. Moreover, which site of measurement can define the presence of osteoporosis/osteopenia (only hip BMD or the three sites?) As far as muscle health is concerned, why did they limit themselves to the European definition of sarcopenia? Furthermore, is the definition of 2010 (EWGSOP1) and/or 2018 (EWGSOP2) being considered? Moreover, this definition must be clearly stated in the protocol, as sarcopenia is a loss of muscle mass coupled with a loss of strength and/or physical performance (different with EWGSOP2, authors to specify). - Animal studies should be part of the exclusion criteria. - What about studies carried out on specific populations such as hospitalized, institutionalized individuals? - What about condition such as osteosarcopenic obesity?

	- To include only the term "osteosarcopenia" in the research strategy seems a bit restrictive to me. In fact, many articles exist whose primary objective is to determine the impact of physical exercise on osteoporosis but evaluate as a secondary objective the effect on muscle mass and muscle strength. - I am a little doubtful about the possible returns from such a research strategy. It seems to me that this strategy is very limited, as there should be very few studies specifically looking at the effect of an intervention on osteosarcopenia as such. In the field of sarcopenia, no clinical trial so far has evaluated the impact of the intervention on the pathology itself but on components of the pathology (i.e., on physical performance only, muscle mass only, ...) and not on sarcopenia as a whole. I think that for osteosarcopenia, we will also find ourselves in this case. The authors should consider it. - If the diagnostic tools used to define sarcopenia differ (e.g. DXA versus BIA for muscle mass, SPPB versus 4m gait speed for physical performance), how will the authors take this into account? - What will be the role of the third reviewer? - Will the extraction be done in duplicate? - Why use two tools to assess the risk of bias? - What will the sensitivity analyses focus on? - The authors do not consider subgroup analyses, meta-regressions? - What will be the effect size considered?

REVIEWER	Vlietstra, Lara University of Otago
REVIEW RETURNED	29-Dec-2020

GENERAL COMMENTS	Overall Good protocol for a systematic review. The results will be of great interest for the osteosarcopenia research area. Please see below for some minor comments. In addition, throughout the manuscript, please consider using the term older adults, rather than elderly people. Furthermore, the tense of the protocol switches a lot from past sentence to future sentence. Please ensure to check the tense of the protocol. Abstract Minor - Page 3, line 21-25: please remove ‘the’ - Page 4, line 25-30: please change osteossarcopenia to osteosarcopenia Introduction Minor - Page 5, line 14-21: please double check the definition of sarcopenia. In the abstract the authors noted that they will include appendicular lean mass, however the current definition as currently written in the introduction is not reflecting the loss of muscle mass. - Page 5, line 23-32: this paragraph needs an introductory sentence. - Page 5, line 39-44: this sentence does not make sense, it is unclear what the authors want to say here.

	Methods Minor - Page 6, line 44-46: what do the authors mean by 'to a grant from the …' ? - Page 6, line 53-58: this sentence is unclear. What do the authors mean by 'even after asking the authors'? Is this part of an exclusion criteria? - Page 9, line 23-25: please be more specific on which sensitivity analysis will be considered.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1
The theme is new and important in terms of public health. However, the research protocol presented here needs further refinement.
Response: Thank you for your carefully assessment of our manuscript and your feedback.

- Title: The World Health Organization recommends that the term 'older adults' be preferred to 'elderly'.
Response: Thank you for raising this important point in relation to the term used. We accept the reviewer's suggestion and we have replaced the term in the full text.

- Abstract: "Improve the osteosarcopenia" does not sound right.
Response: Thank you for your suggestion. We have replaced it by the following sentence:
Page 3; Lines 15 - 17: "…demonstrate the effectiveness of exercises in improving the parameters that lead to the diagnosis of osteosarcopenia…"

- Introduction: "Osteosarcopenia is a recent syndrome", it seems to me rather that the syndrome has been recently defined.
Response: Thank you for your important observation. We have included the following sentence:
Page 5; Line 2: "A recently defined syndrome, osteosarcopenia refers to"

- Introduction: It would be interesting to identify some preliminary studies affirming that physical exercise could have a beneficial effect in the fight against osteosarcopenia.
Response: Thank you for your suggestion. We included a new study (Kemmler et al, 2020) in Line 15 (Page 5) after the following sentence:
"Physical exercise (PE) has been recommended in recent reviews studies as a prevention strategy or non-pharmacological therapeutic approach for osteossarcopenia.".

- Kemmler, W., Kohl, M., Fröhlich, M., Jakob, F., Engelke, K., von Stengel, S., & Schoene, D. (2020). Effects of High-Intensity Resistance Training on Osteopenia and Sarcopenia Parameters in Older Men with Osteosarcopenia—One-Year Results of the Randomized Controlled Franconian Osteopenia and Sarcopenia Trial (FrOST). Journal of Bone and Mineral Research, 35(9), 1634-1644.

- Methods: The definitions of osteoporosis/osteopenia and sarcopenia deserve further clarification. With regard to muscle health, do the authors consider, in the definition of osteosarcopenia, all individuals from the detection of osteopenia or will they make a distinction and a sensitivity analysis with a definition of osteosarcopenia involving all osteopenics and a definition of osteosarcopenia involving only osteoporotics.

Response: Thank you for your review and question this important point. we will consider from the detection of osteopenia. We modified the manuscript with the following sentence:

Page 6; Lines 16 - 18: “We will consider individuals with osteopenia with a T-score < -1 in measurements performed by DEXA on the femoral neck, lumbar spine, or total hip, according to the World Health Organization’s diagnostic criteria for BMD.”

- Methods: Furthermore, is the definition of 2010 (EWGSOP1) and/or 2018 (EWGSOP2) being considered? Moreover, this definition must be clearly stated in the protocol, as sarcopenia is a loss of muscle mass coupled with a loss of strength and/or physical performance (different with EWGSOP2, authors to specify).
Response: Thank you for your important question. Due to the wide variety of definitions and diagnostic criteria for sarcopenia, we decided not to limit these to European consensus. we include the following sentence: Page 6; Lines 19-22:

“Sarcopenia has various definitions and diagnostic criteria. Therefore, the criteria and cutoff points already proposed in studies that perform musculoskeletal measurements will be used, and studies with older adults with reduced muscle strength and mass will be included

- Methods: Animal studies should be part of the exclusion criteria.
Response: Thank you for your suggestion. we agree and include it in the revised version of the manuscript Page 7; Lines 2:

- Methods: What about studies carried out on specific populations such as hospitalized, institutionalized individuals?
Response: Thank you for your important question. Our interest is in the older adults population that lives in the community and long-term institutions. Because of this, studies with hospitalized older adults will be excluded as this population has different characteristics, in addition to underlying disease that could interfere with the use of physical exercise as a treatment for osteosarcopenia. This statement is on the Page 7; Lines 1-2:

“Studies with hospitalized older adults, with a focus on specific conditions, for example, stroke, will be excluded”

- Methods: What about condition such as osteosarcopenic obesity?
Response: Thanks for your important question. Studies with obese older adults will not be excluded, therefore, if found, they will be considered for a systematic review.

- Methods: To include only the term "osteosarcopenia" in the research strategy seems a bit restrictive to me. In fact, many articles exist whose primary objective is to determine the impact of physical exercise on osteoporosis but evaluate as a secondary objective the effect on muscle mass and muscle strength. - I am a little doubtful about the possible returns from such a research strategy. It seems to me that this strategy is very limited, as there should be very few studies specifically looking at the effect of an intervention on osteosarcopenia as such. In the field of sarcopenia, no clinical trial so far has evaluated the impact of the intervention on the pathology itself but on components of the pathology (i.e., on physical performance only, muscle mass only, ...) and not on sarcopenia as a whole. I think that for osteosarcopenia, we will also find ourselves in this case. The authors should consider it.
Response: Thank you for your suggestion. We expanded the terms with synonyms of osteosarcopenia. However, our goal is to analyze the effectiveness of exercise considering osteosarcopenia as a unique syndrome. Because of this, we have restricted the search

strategy so that our objective is achieved. We observed the existence of reviews conducted with the conditions that lead to the diagnosis of osteosarcopenia (osteopenia / osteoporosis and sarcopenia), for example: “Vlietstra L, Hendrickx W, Waters DL. Exercise interventions in healthy older adults with sarcopenia: A systematic review and meta-analysis. Australasian Journal on Ageing. 2018;37:169–83. Available from: http://doi.wiley.com/10.1111/ajag.12521”. However,
considering osteosarcopenia itself, there is still a gap in the literature with studies showing the effectiveness of physical exercise in this syndrome. Therefore, we believe that our review will help to fill this gap.

- Methods: If the diagnostic tools used to define sarcopenia differ (e.g. DXA versus BIA for muscle mass, SPPB versus 4m gait speed for physical performance), how will the authors take this into account?
Response: Thank you for your careful review. All of these methods have good agreement and validity (CRUZ-JENTOFT et al., 2019). We will take into consideration looking to perform a subgroup analysis if it is possible to perform a meta-analysis. In addition, in the tables of the manuscript resulting from this systematic review, the diagnostic methods will be described in detail.

- Methods: What will be the role of the third reviewer?
Response: Thank you for your important question. The third reviewer will be responsible for resolving any disagreements with other reviewers in decisions to include or exclude articles. This statement is on the Page 8; Lines 23:
“Disagreement regarding inclusion will be discussed and resolved by a third reviewer”

- Methods: Will the extraction be done in duplicate?
Response: Thank you for your important question. Data will be extracted by two reviewers (GVS and MN). This statement is in the Page 8; Lines 22-23.

- Methods: Why use two tools to assess the risk of bias?
Response: Thank you for your impotent observation. This was a typo made by us. The two tools evaluate different things. We improve the description for each instrument, as follow:

Page 9; Lines 3 - 9: “Risk of bias will be assessed using the 27-item Downs and Black instrument . A score will be calculated for each manuscript (0–27 points) and scores above 70% will be used to define low risk of bias.
The Grading of Recommendations, Assessment, Development and Evaluation system will be used to assess the quality of evidence. For each research result, the quality of the evidence will be classified as one of the following: high, moderate, low, or very low.”

- What will the sensitivity analyses focus on?
Response: Thank you for your important question. The results will be reported using forest to show the pooled effect of findings and funnel plots to assess precision of the data. Include the following sentence:
Page 9; Lines 18 - 21: "We will group the results in meta-analysis of random effects and these will be reported using a forest plot to show the grouped effect of the findings, in addition, a funnel plot for data accuracy assessment”

- The authors do not consider subgroup analyses, meta-regressions?
Response: Thank you for your important question. We will consider for the analysis of subgroups sex, age group and BMI degree if the data allow to carry out this type of analysis. Include the following sentence: Page 9; Lines 17-18:
“We will consider for the analysis of subgroups sex, age group and BMI”

- What will be the effect size considered?
Response: Thank you for your important question. Include the following sentence: Page 9; Lines 21 – 23.
“The percentage of variability attributable to heterogeneity will be estimated using the I2 test. I2 values of < 40, 40–60, 60–90, and > 90% correspond, respectively, to “not important,” “moderate,” “substantial,” and “considerable” levels of heterogeneity”

Reviewer #2
Good protocol for a systematic review. The results will be of great interest for the osteosarcopenia research area. Please see below for some minor comments. In addition, throughout the manuscript, please consider using the term older adults, rather than elderly people. Furthermore, the tense of the protocol switches a lot from past sentence to future sentence. Please ensure to check the tense of the protocol.
Response: Thank you for raising these important points. We accept all suggestions and include them in the revised version of the manuscript.

- Abstract: Page 3, line 21-25: please remove ‘the’. Page 4, line 25-30: please change osteossarcopenia to osteosarcopenia
Response: Thank you for your suggestion. We accept all suggestions and include them in the revised version of the manuscript.

- Introduction: Page 5, line 14-21: please double check the definition of sarcopenia. In the abstract the authors noted that they will include appendicular lean mass, however the current definition as currently written in the introduction is not reflecting the loss of muscle mass.
Response: Thank you for your concern. We revised the definition and included it in the revised version of the manuscript. We have included the following sentence in the abstract:
Page 3; Lines 15 - 17:“The outcome of this study will be to demonstrate the effectiveness of exercises in improving the parameters that lead to the diagnosis of osteosarcopenia (bone mineral density, quality of muscle mass, muscle strength and physical function) in the older adults”

- Introduction: Page 5, line 23-32: this paragraph needs an introductory sentence.
Response: Thank you for your important observation. We have included the following sentence:
Page 5; Lines 9: “The prevalence of osteosarcopenia varies in different countries.”

-Introduction: Page 5, line 39-44: this sentence does not make sense, it is unclear what the authors want to say here.
Response: Thank you, we agree. We have included the following sentence:
Page 5; Lines 14 - 20: “Physical exercise (PE) has been recommended as a preventive strategy or non-pharmacological therapeutic approach for osteosarcopenia. However, PE is prescribed in the context of isolated conditions, failing to consider osteosarcopenia as a unique condition that encompasses sarcopenia plus osteopenia/osteoporosis. While there is a consensus in the literature on the effectiveness of PE for older adults with sarcopenia and osteoporosis, few studies have explored the efficacy of PE in people with osteosarcopenia..”

-Methods: Page 6, line 53-58: this sentence is unclear. What do the authors mean by ‘even after asking the authors’? Is this part of an exclusion criteria?
Response: Thank you for your important question. The sentence was incomplete and we have rewritten, as follow:
Page xx; Lines xx-xx: “Publications that meet the inclusion criteria but do not have availability of results even after consulting the authors will be excluded.”

- Methods: Page 9, line 23-25: please be more specific on which sensitivity analysis will be considered.
Response: Response: Thank you for your important question. The results will be reported using forest to show the pooled effect of findings and funnel plots to assess precision of the data. Include the following sentence: Page 9; Lines 18 – 21.
"We will group the results in meta-analysis of random effects and these will be reported using a forest plot to show the grouped effect of the findings, in addition, a funnel plot for data accuracy assessment"

VERSION 2 – REVIEW

REVIEWER	Locquet, Médéa University of Liege
REVIEW RETURNED	14-Apr-2021

GENERAL COMMENTS	All comments have been taken into consideration. The paper was already good, it is now very good and will allow for a very enlightening systematic review for researchers working on osteosarcopenia.